Goal: Fast inference for fixed-effects panel quantile regression (FEQR) with common shocks and tens of millions of observations - So basically combine Chiang, Galvao, and Wei (2026) and Lee, Liao, Seo, and Shin (2023).

References I've used but haven't been able to explicitly cite yet include Koenker and Bassett (1978) and Polyak and Juditsky (1992).

Key objects

FEQR objective $Q_{NT}(\alpha, \beta)$ and time-block decomposition $q_t(\alpha, \beta)$

Chiang, Galvao, and Wei (2026) (HTML version) define, in their equation (4), the "standard Koenker-type FEQR estimator (Koenker, 2004) without regularisation" like so:

$$(\hat{\alpha}(\tau), \hat{\beta}(\tau)) = \arg \min_{\{\alpha_i\}_{i=1}^N, \beta} \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} \rho_\tau(Y_{it} - \alpha_i - X_{it}'\beta)\,, \tag{1}$$

"where $\rho_\tau(u) = u\{\tau - \mathbf{1}(u < 0)\}$ is the check loss function" (p. 8) for a fixed quantile index $\tau \in (0, 1)$.[1][2]

Given panel observations $\{(Y_{it}, X_{it}) : i = 1, \dots, N,\ t = 1, \dots, T\}$, individual effects $\alpha = (\alpha_1, \dots, \alpha_N)'$, and common slope $\beta \in \mathbb{R}^p$, define the FEQR objective:

$$Q_{NT}(\alpha, \beta) := \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} \rho_\tau(Y_{it} - \alpha_i - X_{it}'\beta)\,. \tag{2}$$

Alternatively, we can first define the per-time-block objective:

$$q_t(\alpha, \beta) := \frac{1}{N} \sum_{i=1}^{N} \rho_\tau(Y_{it} - \alpha_i - X_{it}'\beta)\,. \tag{3}$$

Grouping terms with a shared t thus lets us rewrite Equation 2 as the equivalent:

$$Q_{NT}(\alpha, \beta) = \frac{1}{T} \sum_{t=1}^{T} q_t(\alpha, \beta). \tag{4}$$

The above formulation of q_t, as an algebraic time-block decomposition of the FEQR objective, follows from p. 3 of Lee, Liao, Seo, and Shin (2023). They first define their cross-sectional population objective:

$$Q(\beta) := E[q(\beta, Y_i)],$$

[1] Note how this is the same FEQR estimator given in equation (2), p. 3 of Galvao and Wang (2015), but with n used for the cross-sectional dimension instead of N.

[2] Also note how the broader longitudinal quantile regression framework comes from Koenker (2004), whereas its large-panel asymptotic theory was developed by Kato, Galvao, and Montes-Rojas (2012).

where the individual loss function $q(\cdot)$ is defined as the check function

$$q(\beta, Y_i) := (y_i - x_i'\beta)\{\tau - \mathbf{1}(y_i - x_i'\beta \leq 0)\}.$$

Their sample estimator is thus:

$$\hat{\beta}_n := \arg\min_{\beta \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^{n} q(\beta, Y_i). \tag{5}$$

My proposed $q_t(\alpha, \beta)$ is, effectively, the panel analog of Lee et al.'s single-observation loss contribution $q(\beta, Y_i)$, except that one time block contains the whole cross section $i = 1, \dots, N$. My $q_t(\cdot)$ also contains an additional $1/N$ normalization. This will eventually make it so that $Q_{NT} = T^{-1} \sum_t q_t$ exactly matches Chiang et al.'s normalization $(NT)^{-1} \sum_{i,t}$, while also ensuring that the β-score per time block is naturally order one in the limit under common shocks.

Qualifier: For the full (α, β), one has to remember that the α_i-coordinate of q_t affects only unit i; the α-subgradient and the β-subgradient live on different effective scales. We'll likely thus forego a direct $(N + p)$-dimensional stochastic approximation theorem and instead use either a profiled slope recursion or an asymptotic equivalence argument for now.

Further reasons for defining the time-block decomposition $q_t(\cdot)$ can be found in Boyd, Mutapcic, and Duchi (2018), who on p. 13 discuss stochastic subgradient methods for large datasets and use an objective of the form

$$f(x) = \frac{1}{m} \sum_{i=1}^{m} F(x; (a_i, b_i)).$$

They then describe the stochastic subgradients obtained by subsampling terms. Instead of the subsampled term being the individual observation (i, t), however, we choose the time block $F_t(\alpha, \beta) = q_t(\alpha, \beta)$ instead. This is because of the DGP assumptions we'll inherit from Chiang et al.: observations with the same t will share B_t under common shocks, meaning the cross-section should be kept together when approximating independent sampling.